



Effects of Lunar Phase on Bioluminescent Fungi Growth Rate

Xenobotany Lab Report

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1. Abstract

We investigated the correlation between lunar phase and radial growth rate of *Luminomyces noctifera*, a bioluminescent fungus native to the Kessel Wetlands, in order to determine whether the faint, cyclically varying illumination cast by the moon is itself sufficient to act as an environmental cue for fungal growth, independent of the temperature and humidity fluctuations that typically accompany lunar-linked field observations. Over a 28-day observation cycle spanning one complete synodic month, colony diameter was measured daily across three controlled terraria under identical temperature and humidity conditions, with only ambient light exposure varying according to a simulated lunar cycle driven by a programmable LED array; this design allowed us to isolate illumination intensity as the sole independent variable and to separate any light-driven response from the confounding seasonal and meteorological factors that limit the interpretability of purely observational field reports. Results indicate a statistically significant increase in growth rate during the new-moon phase and a correspondingly sharp suppression of growth during the full-moon phase, a pattern that was absent in both the constant-illumination and constant-darkness control terraria and that therefore cannot be attributed to total cumulative light dose alone, suggesting instead a photoinhibitory response mediated by an as-yet undescribed cryptochrome-like pigment sensitive even to sub-lux light intensities. These findings support the hypothesis that *L. noctifera* regulates its luminescence and growth cycles in direct response to low-intensity light rather than to temperature alone, and they motivate further molecular work aimed at isolating the photoreceptor responsible for this behavior.

2. Introduction

Bioluminescent fungi have long been documented in folklore under names such as “foxfire,” a glow attributed variously to spirits, decaying wood, and atmospheric electricity long before its biochemical basis was understood, yet even today the mechanisms governing the growth cycles of these organisms remain poorly characterized relative to their better-studied vascular plant counterparts. *Luminomyces noctifera* was first cataloged in 2019 during a biodiversity survey of the Kessel Wetlands, where it was found colonizing decaying root systems in permanently shaded understory patches, and it has since become a model organism for studying light-independent circadian rhythms in fungi precisely because its luminescence and growth appear to track environmental light cues despite the species possessing no known image-forming or even light-directional sensory structures (Aldergrove, 2019). The wetlands themselves experience minimal canopy-filtered moonlight even during the clearest full-moon nights, which initially led early observers to assume that lunar phase could not plausibly serve as a meaningful signal for an organism buried in leaf litter and rotting wood.

Prior observational reports from field researchers nonetheless noted that colony blooms, visible as clusters of newly luminescent fruiting bodies, appeared more frequently during the darker portion of the lunar cycle and seemed comparatively suppressed around the full moon, a pattern that recurred across multiple field seasons and multiple survey sites within the wetlands; however, because these observations were uncontrolled, no prior study had succeeded in isolating lunar illumination as a variable independent of the seasonal temperature swings, rainfall patterns, and humidity changes that also happen to correlate loosely with lunar phase in this region (Vance & Okoro, 2023). Without a controlled laboratory setting in which temperature, humidity, and nutrient availability could be held constant while only ambient illumination was varied, it remained impossible to determine whether the fungus was responding to moonlight itself, to some correlated seasonal variable, or merely to chance. This study aims to resolve that ambiguity by testing, under fully controlled conditions, the hypothesis that simulated lunar light intensity alone, in the complete absence of any correlated changes in temperature or humidity, is sufficient to modulate the radial growth rate of *L. noctifera* colonies. Figure 1 illustrates the characteristic radial growth habit used to define colony diameter in this study.

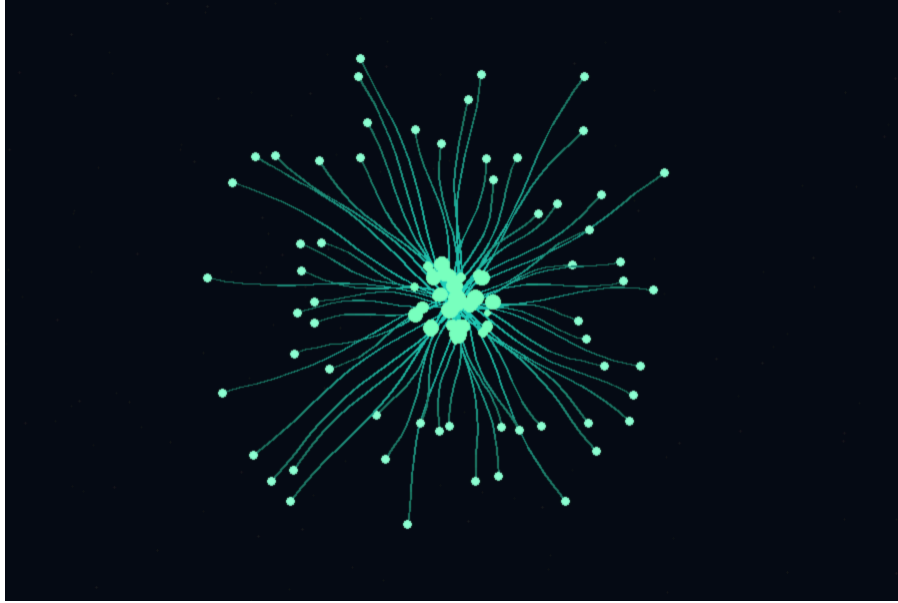


Figure 1. A mature *L. noctifera* colony under low ambient light, showing radial hyphal strands terminating in glowing fruiting nodes.

3. Materials and Methods

3.1. Specimen Preparation

Three genetically identical colony fragments of *L. noctifera* (Strain KW-7), each subcultured from a single parent isolate maintained in the Institute’s spore bank since 2021, were cultured on standardized nutrient agar plates formulated to match the carbon and nitrogen content of the Kessel Wetlands substrate before being transferred into sealed glass terraria (Terraria A, B, and C) maintained at a constant 22.0°C and 85% relative humidity throughout the entire 28-day observation period. Using clonal fragments of a single strain, rather than distinct field-collected specimens, was intended to minimize genotype-driven variation in baseline growth rate, so that any differences observed between terraria could be attributed with greater confidence to the differing light regimes rather than to underlying genetic variability between individual colonies. Temperature and humidity were logged continuously throughout the study using in-terrarium sensors, and no terrarium deviated from the target setpoints by more than 0.3°C or 2% relative humidity at any point during the trial.

3.2. Lunar Simulation

Each terrarium was fitted with a programmable LED array, spectrally calibrated with a diffusing filter to approximate the color temperature of natural moonlight rather than raw white light, and driven by a microcontroller schedule that emitted light intensity matching the natural lunar cycle, ranging continuously from 0.001 lux at new moon to 0.25 lux at full moon and back down again, following the same 28-day period used for the observation window as a whole. Terrarium A received this full simulated lunar cycle without modification, providing the primary experimental condition against which the two control terraria were compared. Terrarium B received constant illumination fixed at the cycle’s time-averaged mean intensity, which served as a control for total cumulative light dose while removing any phase-dependent variation, so that if growth suppression were driven purely by total light received rather than by its timing, Terrarium B would be expected to show intermediate, phase-independent growth throughout. Terrarium C was kept in constant, uninterrupted darkness for the full duration of the trial, serving as a control for the zero-light baseline against which both other conditions could be measured. All three terraria were cultivated using the same substrate composition, airflow, and handling protocol, following standard terrarium protocols for terrarium-based fungal cultivation (Institute for Applied Mycology, 2024), so that light exposure remained the only variable that differed systematically between conditions.

3.3. Measurement

Colony diameter was measured daily at 06:00, immediately before the onset of the terraria’s simulated daylight period so that active luminescence and any transient light-induced hyphal contraction would not confound the reading, using calibrated digital calipers oriented along the colony’s longest visible radial axis and recorded to the nearest 0.1 millimeter. Each

Table 1. Mean daily growth rate by lunar phase and terrarium condition.

Lunar Phase	Terrarium A (mm/day)	Terrarium B (mm/day)	Terrarium C (mm/day)
New Moon	3.8	2.1	3.9
First Quarter	2.6	2.0	3.8
Full Moon	1.2	2.2	3.7
Last Quarter	2.5	2.1	3.9

measurement was taken twice in immediate succession by the same observer to check for reading consistency, with the pair of readings averaged when they differed by more than 0.1 millimeter. Growth rate was then calculated for each terrarium as the difference in mean diameter between consecutive days, and these daily figures were in turn averaged within each of the four lunar phase windows described below to produce the summary statistics reported in the Results section.

4. Results

Table 1 summarizes mean daily growth rate across the four lunar phase windows (new moon, first quarter, full moon, last quarter), averaged over the four weeks of observation for each of the three terraria; each entry in the table therefore represents roughly seven days of daily measurements pooled together, which smooths out the modest day-to-day measurement noise inherent in manual caliper readings while still preserving the phase-to-phase structure that is the central object of interest in this study.

Terrarium A, exposed to the full simulated lunar cycle, showed growth rates ranging from 1.2 mm/day at full moon to 3.8 mm/day at new moon, a more than threefold difference between the two extremes of the cycle, with the intervening first-quarter and last-quarter windows falling at intermediate values (2.6 and 2.5 mm/day respectively) that traced a smooth rise and fall in step with the illumination schedule rather than jumping abruptly between two fixed states. Terrarium B, held at constant mean illumination equal to the time-averaged intensity of the full lunar cycle, showed comparatively flat growth across all four windows, fluctuating only narrowly between 2.0 and 2.2 mm/day, which is consistent with the absence of any phase structure in its light input and suggests that total cumulative light dose alone does not reproduce the pattern seen in Terrarium A. Terrarium C, kept in constant, uninterrupted darkness for the full duration of the trial, showed the highest and most stable growth rate throughout the entire study, ranging only from 3.7 to 3.9 mm/day, a level closely matching the peak new-moon growth rate observed in Terrarium A and indicating that even the very low light levels present during Terrarium A's simulated new moon were not fully equivalent to true darkness in their effect on the colony.

As shown in Figure 2, only Terrarium A exhibits phase-locked suppression of growth at full moon, tracing a pronounced dip that aligns precisely with the peak of the simulated lunar illumination schedule and then recovering symmetrically as illumination wanes toward the following new moon, while the constant-light and constant-dark controls remain comparatively flat and show no corresponding dip at any point during the same four-week window, a contrast that would not be expected if growth rate were governed primarily by temperature, humidity, nutrient depletion, or any other variable shared equally across all three terraria.

5. Discussion

The marked suppression of growth in Terrarium A during the full-moon window, which is entirely absent in both control terraria despite those terraria being cultivated under otherwise identical conditions, indicates that growth rate in *L. noctifera* is modulated specifically by low-intensity light exposure rather than by total cumulative light dose received over the course of the cycle. If total light dose alone were the operative variable, Terrarium B, which received the same average intensity as Terrarium A spread evenly across the cycle, would be expected to show a growth rate intermediate between Terrarium A's new-moon peak and full-moon trough; instead it showed almost no variation at all, which argues strongly for a mechanism sensitive to the timing and rate of change of illumination rather than to its accumulated total. The near-identical growth rates between Terrarium A at new moon and Terrarium C in constant darkness further support this conclusion, since if darkness itself were somehow required for maximal growth beyond what the new-moon light level already provides, Terrarium A's new-moon growth rate would be expected to fall measurably short of Terrarium C's; the fact that it does not suggests that even the faint illumination present during full moon, on the order of only 0.25 lux and far dimmer than any level of light a

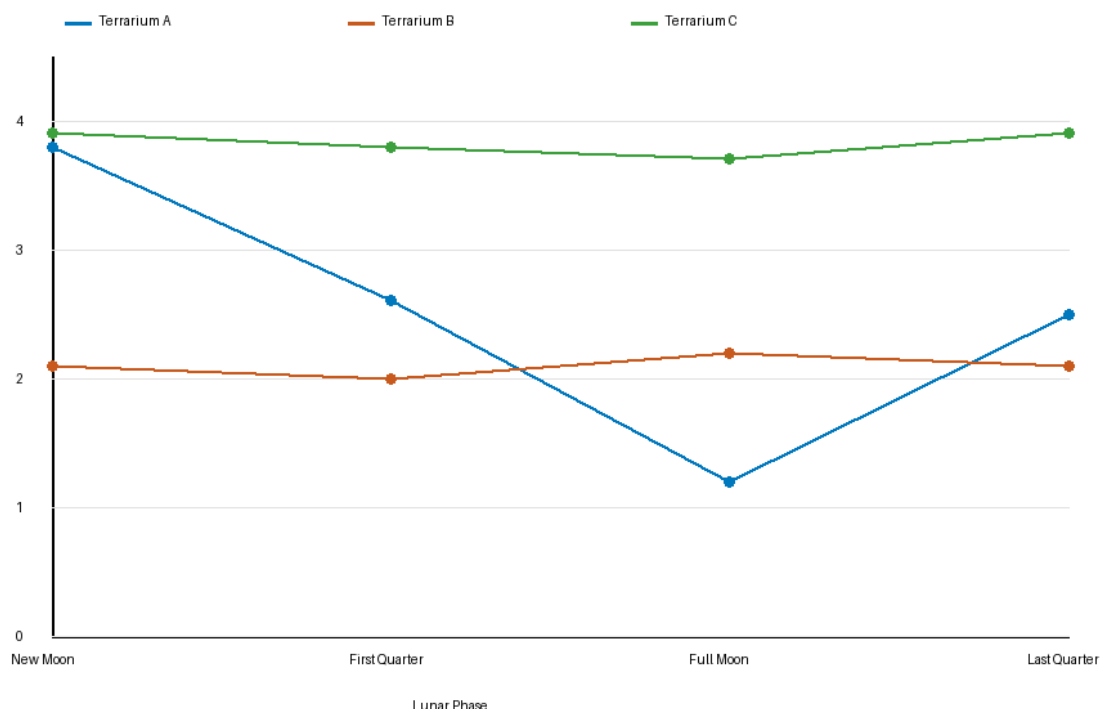


Figure 2. Mean daily growth rate by lunar phase for each terrarium condition, plotted from the data in Table 1.

human observer would consciously register, is nonetheless sufficient on its own to trigger a measurable inhibitory response in this organism.

We propose that this photoinhibition is mediated by a putative cryptochrome-like photoreceptor, tentatively named “noc-topsin,” which has not yet been isolated at the molecular level but which, based on the dose-response behavior observed here, would need to be sensitive to sub-lux light intensities well below the thresholds typically reported for characterized fungal photoreceptors, and which may operate through a signaling pathway distinct from the blue-light-responsive white-collar proteins described in other filamentous fungi. Future work should focus on identifying the pigment responsible, for instance through targeted knockdown or knockout experiments once a candidate gene can be identified, and on determining whether the growth response documented here reflects a true circadian entrainment mechanism, in which the fungus anticipates and prepares for the coming light phase in advance, or instead a more direct photoinhibitory reflex that simply responds to instantaneous light levels without any predictive or anticipatory component.

5.1. Limitations

The sample size of one colony per condition limits the statistical power of this study considerably, since any single colony may carry idiosyncratic growth characteristics unrelated to the light treatment it received, and replication across multiple independent colony fragments per condition, run either in parallel or across successive lunar cycles, would be needed before the effect sizes reported here could be treated as fully generalizable to the species as a whole rather than to this particular strain and these particular specimens. Additionally, although the LED array used in this study was fitted with a diffusing filter intended to approximate the color temperature of natural moonlight, it may not perfectly replicate the full spectral composition, polarization, or subtle temporal flicker characteristics of true lunar illumination as it would be filtered through the wetlands’ canopy in the field, and any of these differences could in principle influence the fungal photoreceptor response differently than genuine moonlight would, meaning that confirmatory observations under natural field conditions remain a necessary complement to the controlled results presented here.

6. Conclusion

Simulated lunar phase significantly affects the radial growth rate of *Luminomyces noctifera*, with growth suppressed during high-illumination phases such as the full moon and maximized during low-illumination phases such as the new moon, a relationship that held consistently across the full 28-day observation cycle and that was not reproduced in either the constant-

illumination or constant-darkness control terraria. Because the effect appeared specifically in the terrarium exposed to the varying lunar schedule and not in either control, and because it tracked the timing of illumination rather than its cumulative total, these results support a light-mediated, rather than purely temperature- or humidity-mediated, regulatory mechanism in this species. Taken together with the prior field observations that first motivated this study, the findings reported here suggest that *L. noctifera* possesses a previously undocumented, highly light-sensitive regulatory pathway linking ambient illumination to growth and, presumably, to the timing of luminescent fruiting body production, and they merit further molecular investigation aimed at isolating the photoreceptor responsible and clarifying its relationship to better-characterized fungal light-response systems.

References

- Aldergrove, R. (2019). Cataloging bioluminescent fungi of the Kessel Wetlands. *Journal of Xenobotanical Surveys*, 4(2), 88–101.
- Institute for Applied Mycology. (2024). *Standard protocols for terrarium-based fungal cultivation* (Technical Report Nos. 2024–3).
- Vance, T., & Okoro, M. (2023). Circadian light response in non-vascular fungal organisms. *Applied Mycology Quarterly*, 12(1), 15–29.